

- Pauli, Wolfgang 257, 258
 biography 373
 neutrinos 263
- Pauli exclusion principle 373
- Pauling, Linus **271**, 373
- Pavel, Andreas 219
- Pavlov, Ivan 225, 243
 biography 373
- Pawer, Georg 80, 82
- Payen, Anselme 190
- Payne, Cecilia 257
- payphone 316
- PCR (polymerase chain reaction) 318, 319
- PDEs 142
- pearl, artificial 228
- pediatrics 36, 119
- pedology 220
- Peking Man 259
- Pelacani, Biagio 68
- pelican, teeth 64
- Pelletier, Pierre-Joseph 181
- Pendry, John 336
- pendulum clock 109, 110
 mercury weight 134
- pendulums 89
- penicillin 262, **272**
 by mouth 274
 first use 263
 Fleming 370
 Florey 370
 structure 275, 280
- Penicillium* 263
- Penydarren* 171
- Penzias, Arno 294
- pepsin 191
- Périer, Florin 106
- periodic table 210, 211, 358
- Perkin, William 373
- Perl, Martin 307
- Permian period 366
- perpetual motion 43, 56, 61
- perpetuum mobile 43
- Perreaux, Louis-Guillaume 210
- Perrier, Carl 270
- Perrine, Charles Dillon 239
- personal computer 185, 302
 IBM-PC 313
 portable 312, 313
- personal portable audio 311
- perspective 68
- Perutz, Max 288
- PET plastic 303
- Peter the Great 134
- Petit, Alexis 180
 biography 373
- Petit, Louis 212
- Petiver, James 131
- petroleum, cracking of 270
- Peurbach, Georg 70
- PGD 323
- pH scale 246
- Sørensen 374
- phage 273
- phagocytosis 222
- Phanerozoic eon 366–67
- phantom limb phenomenon 123
- pharmacopeia, first 33
- Pharos of Alexandria 29
- phase contrast microscope 115
- pheasant 204
- Phillips, William 182
- Philoponus, John 40, 41
- philosophers, ancient Greek 24
- Philosophiae Naturalis Principia Mathematica* 118
- phlogiston theory 124
 disproof 152, 154
 origin 126
- Phobos 217
- Phoenician script 23
- Phoenicians 292
- Phoenix lander 338
- phone, mobile 303
- phonoautograph 207, 218
- phonograph 217, 218–219
- phosphorescence 232, 251
- phosphoric acid, DNA 285
- phosphorus 113, 359
 allotropes 193
- photoelectric effect 224, 235, 242, **255**
- photographic negative 191
- photography 169, 186
 Baekeland 368
 commercial color 243
 Daguerre 369
 digital 306, 307
 first color 208
 first process 192
 Gabor 370
 IR and UV 247
 Muybridge 372
 stop-motion 217
 Talbot 374
 wet plate 200
- photons 251
 and cosmology 345
 and electromagnetic radiation 234, 235
 naming 258, 259
- photoperiodism 271
- photosphere, Sun 54, 55
- photosynthesis 154, **157**, 172
 and chlorophyll 192
 early theories 136
 use of carbon 280
 Villasante 374
- phthisis 36
- phylla, taxonomy 360
- physics, Standard Model 304, 307
- pi 22, 36, **39**, 68, 150
 first use of symbol π 127
 Zhang Heng 374
- Piazzi, Giuseppe 168
- Piccard, Bertrand 347
- Piccard, Jacques 289
- Pick, Arnold 228
- piezoelectric effect 221
- pigeon post 316
- Pillars of Creation 330
- PillCam 330, 331
- Pilltdown skull 249
 hoax 283
- Pinel, Philippe 176
- pinhole camera 61, 86, 95
- Pinta* 71
- Pioneer 10 space probe 303
- Pioneer 11 space probe 305
- pipette 84
- Pisano, Leonardo 58, **59**
- pistol shrimp 268
- piston steam engine 122
- Pithecanthropus* 227
- Pitot, Henri 137
- pitot tube 137
- Pixii, Hippolyte 189
- place notation 58
- plague 66, 67, 103, 106
 bubonic 41
 pneumonic 67
- Planck, Max **236**, 242
 quantum theory 237, 258
- planetarium 65
- planetary motion, laws of 96, 98
- planetary nebula 54
- planetary orbits 37, **100–101**, 364
- planetary years 101, 364
- planets
 extrasolar 328, 346, 347
 reference tables 364
- Plank, Max 373
- Plank satellite 349
- plants
 Banks 368
 classification 13, 30, 89, 126, 160, 360–63
 domestication 13
 first genome sequenced 337
 genetic engineering of 320, 321
 Hales 371
 physiology 371
 Plumier 126
 reproduction **189**
- plasma, in stars 54, 55
- plastics
 Bakelite 368
 first synthetic 243
 PET 303
 story of **260–61**
- platinum 359
 early mention 82
- Plato **25**, 28
 and geometry 26, 27
 biography 373
- Platonic solids 26, 27, 29
- Platter, Felix 89
- platypus 167
- player piano 218
- Playfair, John 169
- PlayStation 3 342
- plesiosaur 176, 182
- Pliny the Elder 36
- Plotinus 38
- plough 15
 fire 12
 Rotherham 136, 137
- Plücker, Julius 202
- plum pudding model of the atom 239
- Plumier, Charles 126
- Plunket, Roy 261
- pluripotent cells 312
- Pluto 263
 dwarf planet 342
 first photograph 252
- plutonium 272, 359
- pneumatic tire 21, 225
- removable 227
- pneumatics 30
- pneumonic plague 67
- Pockels, Agnes 227
- pocket calculator 185
- pocket watch 72
- Poincaré, Henri **227**, 373
- poisoning, heavy metal 286
- poisons 57
- Poisson, Denis 192
- Poisson distribution 192
- polarization **179**, 182
- polarized light 178
- polarizing light microscope 115
- polio vaccine 282, 283
- poliomyelitis 246
- pollen 123
 for dating 253
 pollination 88
- pollen tube 188, 189
- pollination 88
- polonium 233, 266, 267, 359
- Polyakov, Valeri 330
- polyester 268
- polyethylene 261
- polyhedra 26
- polymerase chain reaction (PCR) 318, 319
- polymers 261
- polypropylene 261
- polystyrene 261, 272, 273
- polyvinyl chloride (PVC) 215, 258, 260, 261
- Pong video game 303
- Pons, Stanley 323
- Pontecorvo, Bruno 296
- Pool, Judith 294
- Popper, Erwin 246
- population genetics 246
- population growth 303
 challenges 349
 Malthus 166
- porcelain 42, 128
- portable digital music player, iPod 338
- Porter, Rodney 291
- portolan chart 132
- Posidonios of Apamea 32
- positron 265
- postal service 316
- potassium 267, 359
 Davy 369
 isolation 173
- Potrykus, Ingo 337
- potter's wheel 20, 21
- pottery 13
- Pouillet, Claude 193
- Poulsen, Valdemar 219
- pound weight 64, 84
- power
 equations 357
 units of measurement 352, 353, 354
- power loom 156, 157
- Pravaz, Charles 201
- precession 31
- preimplantation genetic diagnosis 323
- preservation, anatomical **79**
- pressure
 equations 356
 units of measurement 352, 353, 354
- water **92**
- pressure cooker 117, 122
- pressure in fluids 368
- Prestwich, Joseph 206, 207
- Prévost, Pierre 161
- Priestley, Joseph 150, 151, 152
 biography 373
- primates 362
- Prime Meridian 223
- prime number theorem 163
- prime numbers 31, 143
 Mersenne 152
- Principia* (Newton) 118
- printed book, first 47
- printing 69
 linotype 224
 movable type 51, 69, 373
 stereotype 135
 wood block 40, **41**, 56
- printing press 178
- prions 314
- prisms 112
- Pritchard, Thomas 154
- probability theory 107, 110, 122, 129
- procaine 242
- Proclus 39
- Procopius 40, 41
- products of chemical reactions 174
- programming cells **343**
- Project Dina 278
- Prokhorov, Alexander 283
- promethium 274, 359
 discovery 278
- prominence, solar 55
- promissory notes 46
- prontosil 268
- Prorospatharius, Theophilus 42
- prosthesis
 hand 83
 intelligent 328, 329
- protactinium 266, 359
- proteins
 amino acids 243, **281**
 DNA 285
 naming 193
 structure 373
- Proterozoic eon 366
- protists 210
- proto-Canaanite script 23
- proto-Sinaitic script 23
- proton linear accelerator 278, 279
- protons 178, 250, 251, 253, 266, 267
 and cosmology 344
 and Rutherford 252
 naming 254, 255
 Standard Model 304
- protostars 54
- Proust, Joseph-Louis 373
- Proust's law 373
- Prout, William 178, 187
- psi particle 304
- psychoanalysis 228
 Freud 370
- psychology 226
- pterosaur 187
- PTFE 261
- Ptolemy (Claudius Ptolemaeus) of Alexandria 28, 37, 71
 biography 373
- public health 66
- Puckle, James 131
- Puckle gun 131
- puerperal fever 110
- Puffing Billy 178
- Puffing Devil 168
- pulley, simple machine 34, 35
- pulmonary circulation **81**
- pulsar 296
 double 342
- pulse watch 128
- pump, Ctesibian 30
- punched cards 224
- pupil reflex 143
- Purcel, Edward 278
- push-button telephone 291
- PVC (polyvinyl chloride) 215, 258, 260, 261
- Pylari, Giacomo 131
- pyramids 18, 19
- pyréolophore 173
- Pythagoras 25, 26
 biography 373
- Pythagoras's theorem 22, 24, **25**, 373

Q

- Qiuqian, Zhang 40
- quadrant, instrument 71, 130, 133
- quadratic equations 22, 38, 46, 93
- quagga 222
- quanta, coining of term 237
- quantum computing 309, 312, 333
- quantum electrodynamics 279, 370
- quantum mechanics 248, 258, 259, 368–69, 373
- quantum physics 251
- Planck 236
- quantum theory **258**, 259
- quarantine 66, 67
- Quare, Daniel 123
- quark 244, 291, 296, 357
 charm 304
- quartz clock 109, 259
- quasar 294, 310
- Quaternary Period 188
- qubit 330
- quinine 103, 181, 274